CardioGraph AI

Autonomous Coronary Vascular Network & Plaque Rupture Predictor

Target Event: WeMakeDevs Graph Hacks (FalkorDB) | **Scientific Vision:** Demis Hassabis (DeepMind) & Dario Amodei (Anthropic) | **Version:** 1.0 Final

---

1. Executive Summary & Scientific Origin

Cardiovascular disease is the **#1 cause of death globally**, claiming 18 million lives each year. In late 2025, Nobel Laureate **Sir Demis Hassabis** and Google DeepMind used AlphaFold to resolve the 3D molecular structure of **ApoB100**, the structural backbone of LDL cholesterol. Concurrently, **Dario Amodei** (CEO of Anthropic, Ph.D. in Biophysics) identified the causal modeling of complex cardiovascular and biological networks as AI's most critical unsolved medical frontier.

While AlphaFold mapped static protein geometry, medicine remains blind to the living dynamic: **how vulnerable plaques destabilize at turbulent arterial branching junctions and trigger sudden fatal heart attacks**. **CardioGraph AI** introduces an autonomous clinical agent powered by **FalkorDB** that models the patient's coronary arterial tree as a graph, evaluates hemodynamic turbulence at bifurcations, and computes ischemic muscle death blast radius before clinical catastrophe occurs.

> ***The Core Breakthrough:*** *Moving beyond flat cholesterol blood tests and 2D angiograms by modeling the coronary arterial tree as a living, directed flow graph with FalkorDB.*

2. Clinical Users & Economic Value

Target User	Clinical Setting	Daily Unsolved Problem & Economic Impact
Interventional Cardiologists	Cardiac Cath-Lab	Deciding whether a 30-50% non-obstructive plaque is stable or on the verge of fatal rupture. Avoids unnecessary $10,000 stents while catching high-risk soft lesions.
Cardiac Radiologists	CT Angiography (CCTA)	Tracing 15+ coronary branches across 500+ CT slices manually takes 45 mins. CardioGraph provides an instant, auto-generated interactive vascular graph.
Emergency Room Physicians	Chest Pain Triage	Differentiating harmless non-cardiac chest pain from impending acute myocardial infarction using graph perfusion risk.
Hospitals & Insurers	Health Systems / Payers	Treating a massive heart attack and ICU stay costs $50,000–$150,000. Preventing acute events saves millions per healthcare network.

3. The Clinical Reality: 'The Rupture Paradox'

Standard cardiology relies on 'stenosis percentage'. A 70% or 80% blockage is stented because it looks severe. However, **clinical studies prove over 65% of fatal heart attacks occur at plaques with LESS than 50% stenosis**. These 'vulnerable plaques' are soft, lipid-rich cores covered by fragile fibrous caps. When situated near arterial bifurcations, hemodynamic shear turbulence tears the cap, triggering sudden massive thrombosis (the 'Widow Maker'). Neither SQL databases nor vector embeddings can model recursive downstream flow networks and fluid turbulence.

4. FalkorDB Vascular Network Graph Schema

CardioGraph AI structures patient coronary anatomy and biology into a multi-hop graph:

Node Definitions:

- **ArterySegment**: name (LAD_Proximal, LAD_Mid, LCx, RCA, Diagonal_1), diameter_mm, length_mm, flow_rate.
- **Bifurcation**: id, branch_angle_deg, shear_stress_index, turbulence_factor.
- **Plaque**: id, stenosis_pct (e.g. 38%), plaque_type ('soft_lipid' vs 'fibrocalcific'), cap_thickness_um, vulnerability_score.
- **BiomarkerProfile**: apob_mg_dl, lpa_nmol_l, hscrp_mg_l, troponin_i.
- **MyocardiumZone**: zone_name ('Anterior_Apex', 'Septum', 'Inferior_Wall'), mass_percentage.

Edge Relationships:

- **(:ArterySegment)-[:FLOWS_INTO {resistance: Float}]->(:ArterySegment)** — Models blood flow topology.
- **(:ArterySegment)-[:CONTAINS_PLAQUE]->(:Plaque)** — Pinpoints exact anatomical lesion location.
- **(:Plaque)-[:ADJACENT_TO {distance_mm: Float}]->(:Bifurcation)** — Captures high-turbulent physical stress.
- **(:BiomarkerProfile)-[:DESTABILIZES {hazard_multiplier: Float}]->(:Plaque)** — Biological cap erosion.
- **(:ArterySegment)-[:PERFUSES]->(:MyocardiumZone)** — Maps arterial blood supply to living heart muscle tissue.

5. FalkorDB Native Graph Algorithms as Agent Tools

1. **Downstream Ischemic Blast Radius (shortestPath & Reachability)**: Traces every downstream path from an unstable plaque. If a clot forms at LAD_Mid, the agent calculates that 45% of the anterior and septal myocardium will starve of oxygen within 30 minutes.

2. **Critical Bifurcation Chokepoints (Betweenness Centrality)**: Computes arterial junctions sustaining the greatest systemic flow volume. Pinpoints single points of failure (SPOF) in the patient's coronary tree.

3. **Multi-Hop Vulnerability Detection**: Traverses (Biomarkers) -> (Plaque) -> (Bifurcation) to identify non-obstructive plaques (<50%) that have high biological erosion and elevated physical shear strain.

6. Architecture & Hackathon Execution Plan

Component	Technology	Hackathon Functionality
Graph Context Engine	FalkorDB (Docker / Port 6379)	GraphBLAS linear algebra engine storing patient vascular trees with sub-millisecond multi-hop Cypher queries.
Agent Framework & MCP	Python / FastAPI / LangGraph	Model Context Protocol (MCP) server exposing vascular query tools directly to LLMs (Claude 3.5 / GPT-4o).
Clinical Frontend	React / Vite / Cytoscape.js	Interactive 2D coronary arterial tree. When the agent warns of rupture, the branch pulses red and ischemic zones glow.

Deliverables Ready: Full specification, Cypher schema, and prototype modules prepared in d:\MIT\CardioGraph-AI and D:\CardioGraph AI.